

HYDROGENATED NBR SYNTHESIS AND PROCESS DEVELOPMENT: CATALYTIC HYDROGENATION, LOW-VISCOSITY ROUTES, AND FUNCTIONALIZED HNBR LATEX PROCESSING

1. ABSTRACT

This report synthesizes six patents relevant to hydrogenated nitrile butadiene rubber (HNBR), focusing on how NBR precursors are converted to partially or fully hydrogenated products and how process choices affect residual unsaturation, viscosity, and downstream performance. The target material is HNBR derived from acrylonitrile/butadiene copolymers, including conventional NBR, low-molecular-weight HNBR, and functionalized HNBR terpolymers. Across the patent set, the dominant synthesis theme is selective hydrogenation of diene unsaturation in solution or latex, typically using noble-metal catalysts from groups 8–10, with strong emphasis on catalyst selectivity, catalyst removal, and control of gel formation or aging stability. EP3248988B1 is centered on homogeneous rhodium-catalyzed hydrogenation of nitrile groups and tandem hydrogenation in solution, using $\text{RhH}(\text{P}i\text{Pr}_3)_3$ with phosphine additives to suppress side reactions and gel formation. US20210340285A1 addresses HNBR production with ruthenium, palladium, or rhodium compounds at low ppm catalyst levels, aiming to improve aging properties by minimizing residual noble metals and avoiding post-hydrogenation purification. US7919563B2 and US7772328B2 pursue a different route: metathesis-based molecular-weight reduction of nitrile rubber prior to or in conjunction with hydrogenation, yielding low-Mooney HNBR with narrower molecular-weight distributions. US11827726B2 focuses on latex hydrogenation of PEG acrylate-functionalized NBR followed by coagulation, combining HNBR formation with retained functional comonomer content to improve processability and tailor properties. US10731020B2 is not a direct HNBR synthesis patent, but it is relevant as it explicitly includes hydrogenated nitrile rubber among base rubbers for vulcanization-molding compositions and provides processing constraints that reflect the importance of Mooney viscosity control in HNBR applications. Overall, the landscape shows two major process directions: direct catalytic hydrogenation of NBR/HNBR precursors in solution or latex, and precursor modification by metathesis or functional comonomer incorporation to manage viscosity and application performance.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: EP3248988B1
- Patent Title: Hydrogenation of nitrile groups in hydrogenated nitrile butadiene rubber and ...
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2020
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses solution hydrogenation of hydrogenated acrylonitrile butadiene rubber (HNBR) nitrile groups and tandem hydrogenation of nitrile and olefinic groups in NBR. The process uses an unsupported rhodium catalyst, specifically $\text{RhH}(\text{PiPr}_3)_3$, together with an additive such as triphenylphosphine to improve selectivity and suppress gel formation. - Nitrile-group monomer content: 10 to 60 % by weight — Detailed description - Additional monomer content: 0 to 40 % by weight — Detailed description - Mooney viscosity ($\text{ML1}+4$ @ 100 °C): 20 to 250 — Detailed description - Degree of hydrogenation of diene units: 50 % to 100 % — Detailed description - Catalyst identity: tris-(triisopropylphosphine)-hydrido-rhodium, $\text{RhH}(\text{PiPr}_3)_3$ — Claims - Additive: triphenylphosphine (TPP) — Claims

Relevant Experimental Evidence - Example block: the patent explains why polymer nitrile hydrogenation is difficult and why homogeneous catalysts are preferred to avoid gel formation. - Example block: it contrasts small-molecule nitrile hydrogenation with polymer systems and highlights selectivity limitations of heterogeneous catalysts. - Example block: it demonstrates the intended use of $\text{RhH}(\text{PiPr}_3)_3$ with phosphine additives for high-selectivity hydrogenation in solution.

Technical Relevance This is a core HNBR synthesis patent because it directly targets hydrogenation chemistry in NBR/HNBR polymers. It is especially relevant for selective hydrogenation control, gel suppression, and homogeneous rhodium catalysis in solution.

Patent 2

- Patent Number: US7919563B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2011
- Legal Status: Unknown

- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a metathesis-based route to low molecular weight hydrogenated nitrile rubber. A nitrile rubber is subjected to metathesis in the presence of Ru/Os catalysts and optionally a co-olefin, producing HNBR with reduced molecular weight, lower Mooney viscosity, and narrower molecular-weight distribution. - Molecular weight (MW): 30,000 to 250,000 — Claims - Mooney viscosity (ML 1+4 @ 100 deg. C.): 3 to 50 — Claims - Polydispersity index: less than 2.5 — Claims - Molecular weight (Mw): 40,000 to 220,000 — Claims - Polydispersity index: less than 2.3 — Claims - Cement concentration: 15 wt.-% solid — Example 1 - MCB amount: 1133 g — Example 1 - Ethylene pressure: 6.9×10^5 Pa — Example 1 - Degassing cycles: 3 — Example 1 - Cement concentration: 6 wt.-% solid — Example 3 - 1-hexene amount: 150 g — Example 4 - Nitrogen degassing: 3 times — Example 4 - Reactor temperature: 80° C. — Table 1 - Catalyst load: 0.05 phr — Table 1 - Catalyst load: 0.10 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Mn: 27 kg/mol — Table 2 - Mw: 54 kg/mol — Table 2 - PDI: 2.0 — Table 2 - Mooney viscosity: 43 — Table 2 - Mw reduction: up to fivefold — Table 2 - PDI minimum: 1.90 — Table 2

Relevant Experimental Evidence - Example 1: ethylene co-olefin metathesis was run on a 15 wt.-% cement after repeated degassing under 6.9×10^5 Pa ethylene. - Example 2: a second 15 wt.-% cement experiment repeated the ethylene-degassing protocol to compare catalyst loading effects. - Example 3: a 6 wt.-% cement experiment showed stronger molecular-weight reduction and lower Mooney viscosity. - Example 4: 1-hexene was used as co-olefin under nitrogen degassing, demonstrating an alternative chain-transfer/metathesis route. - Example 4/Table 2: the product properties showed reduced Mn/Mw and narrowed PDI relative to the starting polymer.

Technical Relevance This patent is highly relevant because it provides an alternative route to HNBR processability improvement by lowering molecular weight before or during hydrogenation-related processing. The disclosed low-Mooney, narrow-PDI HNBR is directly useful for applications where conventional HNBR is too viscous.

Patent 3

- Patent Number: US7772328B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent is essentially the same low-molecular-weight HNBR metathesis concept as US7919563B2, disclosing preparation of

hydrogenated nitrile rubber with reduced molecular weight and narrower molecular-weight distribution. It uses metathesis of nitrile rubber with a co-olefin in the presence of Ru/Os catalysts to improve processability. - Molecular weight (MW): 30,000 to 250,000 — Claims - Mooney viscosity (ML 1+4 @100 deg. C.): less than 10 — Claims - Polydispersity index: less than 2.5 — Claims - Cement concentration: 15 wt.-% solid — Example 1 - MCB amount: 1133 g — Example 1 - Ethylene pressure: 6.9×10^5 Pa — Example 1 - Degassing cycles: 3 — Example 1 - Cement concentration: 6 wt.-% solid — Example 3 - 1-hexene amount: 150 g — Example 4 - Nitrogen degassing: 3 times — Example 4 - Reactor temperature: 80° C. — Table 1 - Catalyst load: 0.05 phr — Table 1 - Catalyst load: 0.10 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Mn: 27 kg/mol — Table 2 - Mw: 54 kg/mol — Table 2 - PDI: 2.0 — Table 2 - Mooney viscosity: 43 — Table 2 - Mw reduction: up to fivefold — Table 2 - PDI minimum: 1.90 — Table 2

Relevant Experimental Evidence - Example 1: ethylene was used as co-olefin with repeated degassing of a 15 wt.-% cement. - Example 2: a second ethylene-based run confirmed the effect of catalyst loading on molecular-weight reduction. - Example 3: a lower-solids cement gave a more pronounced reduction in molecular weight and Mooney viscosity. - Example 4: 1-hexene was used as co-olefin, showing that alternative olefins can be used in the metathesis step. - Table 2: the product exhibited substantially lower Mw and narrower PDI than the substrate.

Technical Relevance This patent is directly relevant to HNBR synthesis because it defines a route to low-viscosity hydrogenated nitrile rubber, a major processing challenge in HNBR manufacturing. It complements direct hydrogenation patents by addressing precursor molecular-weight control and final rheology.

Patent 4

- Patent Number: US20210340285A1
- Patent Title: Method for producing hydrogenated nitrile rubber and hnbr compositions thereof
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses hydrogenation of at least partially unsaturated nitrile rubber in solution using ruthenium, palladium, or rhodium compounds at low ppm levels, with the goal of producing HNBR that has improved aging properties. The emphasis is on minimizing catalyst residues and avoiding a separate catalyst-removal step after hydrogenation. - Conjugated diene content: 40% to 90% by weight — Claims - α,β -unsaturated nitrile content: 10% to 60% by weight — Claims - Ruthenium amount: 10 ppm to 200 ppm — Claims - Ruthenium amount: 10 ppm to 150 ppm — Claims - Ruthenium amount: 10 ppm to 120 ppm — Claims - Ruthenium amount: 10 ppm to 79 ppm — Claims -

Ruthenium amount: 42 ppm to 79 ppm — Claims - Palladium amount: 20 ppm to <67 ppm — Claims - Palladium amount: 44 ppm to <67 ppm — Claims - Rhodium amount: 50 ppm to <270 ppm — Claims - Rhodium amount: 79 ppm to <270 ppm — Claims - Rhodium amount: 50 ppm to 250 ppm — Claims - Rhodium amount: 50 ppm to 170 ppm — Claims - Rhodium amount: 79 ppm to 170 ppm — Claims - Hydrogenation temperature: 60 to 200° C. — Claims - Hydrogenation pressure: 700000 pascals to 15000000 pascals — Claims - Hydrogenation time: 1 to 24 hours — Claims - Ageing test temperature: 140° C. — Detailed description - Ageing test duration: 4 days — Detailed description - Gel content increase: less than 10% — Detailed description - Mooney viscosity increase: less than 40% — Detailed description

Relevant Experimental Evidence - Example block: the patent reviews prior art catalyst-removal methods such as methanol washing, adsorption, ultrafiltration, ion exchange, and ionic liquids. - Example block: it identifies catalyst residues as a cause of dark discoloration, poor aging, and higher gel content in HNBR. - Example block: the invention aims to produce HNBR with good aging properties without a separate noble-metal removal step.

Technical Relevance This patent is highly relevant because it is a direct HNBR production method based on catalytic hydrogenation of NBR in solution. Its low-ppm noble-metal strategy and aging-performance targets are important for industrial HNBR quality control.

Patent 5

- Patent Number: US11827726B2
- Patent Title: Process for producing PEG acrylate-HNBR copolymer
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2023
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses latex hydrogenation of PEG acrylate-functionalized NBR followed by coagulation to produce PEG acrylate-HNBR copolymer. The process combines selective hydrogenation of the diene units with retention of PEG acrylate functionality to improve processability and tailor final properties. - α,β -ethylenically unsaturated nitrile unit: 10% to 60% by weight — Claims - Conjugated diene unit: 20% to 80% by weight — Claims - PEG acrylate unit: 30% to 60% by weight — Claims - α,β -ethylenically unsaturated nitrile unit: 15% to 40% by weight — Claims - Conjugated diene unit: 30% to 65% by weight — Claims - PEG acrylate unit: 30% to 50% by weight — Claims - α,β -ethylenically unsaturated nitrile unit: 15% to 35% by weight — Claims - Conjugated diene unit: 35% to 60% by weight — Claims - PEG acrylate unit: 30% to 50% by weight — Claims - PEG acrylate formula parameter n: 1 to 8 — Claims - PEG acrylate formula parameter R1: hydrogen or methyl — Claims - PEG acrylate formula parameter R: hydrogen or branched or unbranched C1-C20-alkyl — Claims - Reaction temperature for latex hydrogenation: 80° C. to 160° C. —

Background - Hydrogen pressure for latex hydrogenation: 3000000 to 8000000 pascals —

Background - Transition metal catalyst group: 8, 9 or 10 — Background

Relevant Experimental Evidence - Example block: the patent explains that direct latex hydrogenation of NBR is known and gives typical reaction conditions. - Example block: it contrasts latex hydrogenation with solution hydrogenation and metathesis-based molecular-weight reduction. - Example block: it motivates PEG acrylate functionalization as a route to improved processability in HNBR terpolymers.

Technical Relevance This patent is relevant because it extends HNBR synthesis to functionalized PEG acrylate-containing copolymers, not just conventional HNBR. It provides a latex-hydrogenation route that is industrially attractive for processability and property tuning.

Patent 6

- Patent Number: US10731020B2
- Patent Title: Rubber composition for vulcanization molding, process for manufacturing the ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2020
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent is not a direct HNBR synthesis patent; it discloses rubber compositions for vulcanization molding and explicitly includes hydrogenated nitrile rubber among the base rubbers. Its relevance to HNBR lies in downstream processing, where Mooney viscosity and filler/hollow-particle handling determine whether HNBR-based compounds can be molded successfully. - Further expansion ratio of hollow particles: 20 to 80% — Claims - True specific gravity d_1 of hollow particles: 0.02 to 0.40 — Claims - Base rubber Mooney viscosity ML (1+4) at 100° C.: 5 to 90 — Claims - Rubber composition hollow particle amount: 0.1 to 30 wt % — Detailed description - Rubber composition base rubber amount: 10 to 95 wt % — Detailed description - Escape-initiation temperature of blowing agent: 110 to 350° C. — Detailed description - Mean particle size of hollow particles: 15 to 100 μm — Detailed description - Example 1 base rubber Mooney viscosity ML (1+4) at 100° C.: 54 — Example 1 - Example 1 hollow particle amount: 2.05 parts by weight — Example 1 - Example 1 process oil: 20 parts by weight — Example 1 - Example 1 talc: 100 parts by weight — Example 1 - Example 1 vulcanization accelerator: 3.5 parts by weight — Example 1 - Example 1 sulfur: 0.4 parts by weight — Example 1 - Example 1 kneading temperature: 60° C. — Example 1 - Example 1 kneading time: about 10 minutes — Example 1 - Example 1 vulcanization molding temperature: 140° C. — Example 1 - Example 1 true specific gravity d_3 : 0.97 — Example 1 - Example 1 theoretical true specific gravity d_5 : 0.96 — Example 1 - Example 1 breakage ratio: 3.71% — Example 1 - Example 3 vulcanization temperature: 150° C. — Example 3 - Example 3 vulcanization time: 15 minutes — Example 3 - Example 6 cylinder temperature: 180° C. —

Example 6 - Example 6 die temperature: 60° C. — Example 6 - Example 6 injection pressure: 15 MPa — Example 6 - Example 11 hollow particle mean particle size D50: 129.8 µm — Example 11

Relevant Experimental Evidence - Example 1: a composition with base rubber Mooney 54 and 2.05 parts hollow particles gave low breakage and good lightweight performance. - Example 3: extrusion and vulcanization at 150° C. for 15 minutes produced a lightweight molded product with good dimensional stability. - Example 6: injection molding at 180° C., 60° C. die temperature, and 15 MPa gave a molded product with good appearance and stability. - Example 11: larger hollow particles (D50 129.8 µm) still produced an acceptable molded product with only minor surface unevenness. - Comparative examples: inappropriate hollow-particle expansion ratios or high base-rubber viscosity caused breakage and poor lightweight effect.

Technical Relevance Although not a synthesis patent, it is relevant to HNBR because it explicitly includes hydrogenated nitrile rubber as a base rubber in molding compositions. It highlights the importance of Mooney viscosity and filler compatibility for HNBR processing and end-use performance.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- EP3248988B1 and US20210340285A1 both use noble-metal catalysis in solution, but EP3248988B1 emphasizes unsupported rhodium with phosphine additive for selective nitrile hydrogenation, whereas US20210340285A1 emphasizes low-ppm Ru/Pd/Rh loading and catalyst-residue control for aging stability.
- US7919563B2 and US7772328B2 both use metathesis to reduce molecular weight of nitrile rubber before/while making HNBR; the disclosed examples show ethylene or 1-hexene co-olefins, 80° C. reactor temperature, and catalyst loadings of 0.05–0.25 phr.
- The metathesis patents target low Mooney viscosity and narrow PDI, while the direct hydrogenation patents target high degree of hydrogenation, low residual unsaturation, and suppression of gel formation.
- US11827726B2 differs from the other HNBR patents by hydrogenating PEG acrylate-functionalized NBR latex rather than conventional solution polymer; it preserves functional comonomer content while converting diene unsaturation.
- US20210340285A1 and EP3248988B1 both address catalyst selectivity and side reactions, but US20210340285A1 is more focused on post-hydrogenation aging and noble-metal residue management.
- US10731020B2 is downstream-processing oriented rather than synthetic; it shows that HNBR application performance is strongly influenced by base-rubber Mooney viscosity and compound formulation, not only by hydrogenation chemistry.

4. CONCLUSION

The evidence supports two principal HNBR synthesis strategies: direct catalytic hydrogenation of NBR/HNBR precursors in solution or latex, and precursor modification by metathesis to lower molecular weight and improve processability. For direct hydrogenation, the most relevant parameters are selective noble-metal catalysis, solution or latex processing, hydrogenation temperatures around 60 to 200° C., pressures in the MPa range, and careful control of catalyst residues to avoid gel formation and aging loss. For processability-focused HNBR, the metathesis patents show that low Mooney viscosity and narrow molecular-weight distribution can be achieved using Ru/Os catalysts with co-olefins such as ethylene or 1-hexene at about 80° C. and low phr catalyst loadings. Functionalized latex hydrogenation, as in PEG acrylate-HNBR, broadens the design space for specialty elastomers. Based on the evidence, the most robust synthesis targets are: selective hydrogenation in solution or latex, low residual unsaturation, minimal noble-metal contamination, and precursor molecular-weight control to keep final Mooney viscosity in a processable range.

5. REFERENCES

- EP3248988B1 | Hydrogenation of nitrile groups in hydrogenated nitrile butadiene rubber and ... | Not disclosed | EP | 2020 | <https://patents.google.com/patent/EP3248988B1/en>
- US7919563B2 | Low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2011 | <https://patents.google.com/patent/US7919563B2/en>
- US7772328B2 | Low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2010 | <https://patents.google.com/patent/US7772328B2/en>
- US20210340285A1 | Method for producing hydrogenated nitrile rubber and hnbr compositions thereof | Not disclosed | US | 2021 | <https://patents.google.com/patent/US20210340285A1/en>
- US11827726B2 | Process for producing PEG acrylate-HNBR copolymer | Not disclosed | US | 2023 | <https://patents.google.com/patent/US11827726B2/en>
- US10731020B2 | Rubber composition for vulcanization molding, process for manufacturing the ... | Not disclosed | US | 2020 | <https://patents.google.com/patent/US10731020B2/en>